

TerraStay Hotel Booking System

Developer Reference — Flows & Database Design

Project:
Stack:
Repo:
Login:

Hotel Inventory & Booking System
Laravel 13 · PHP 8.3 · Sanctum · Blade · MySQL/SQLite · Redis
BookingApp (hotel-booking-temp)
admin@terrastay.com / password

1. What This Project Is

A hotel inventory and **date-aware availability search** system with:

- **Web UI** — session auth, Blade + Bootstrap 5
- **JSON API** — Sanctum token auth
- **Shared service layer** — business logic is never duplicated between web and API

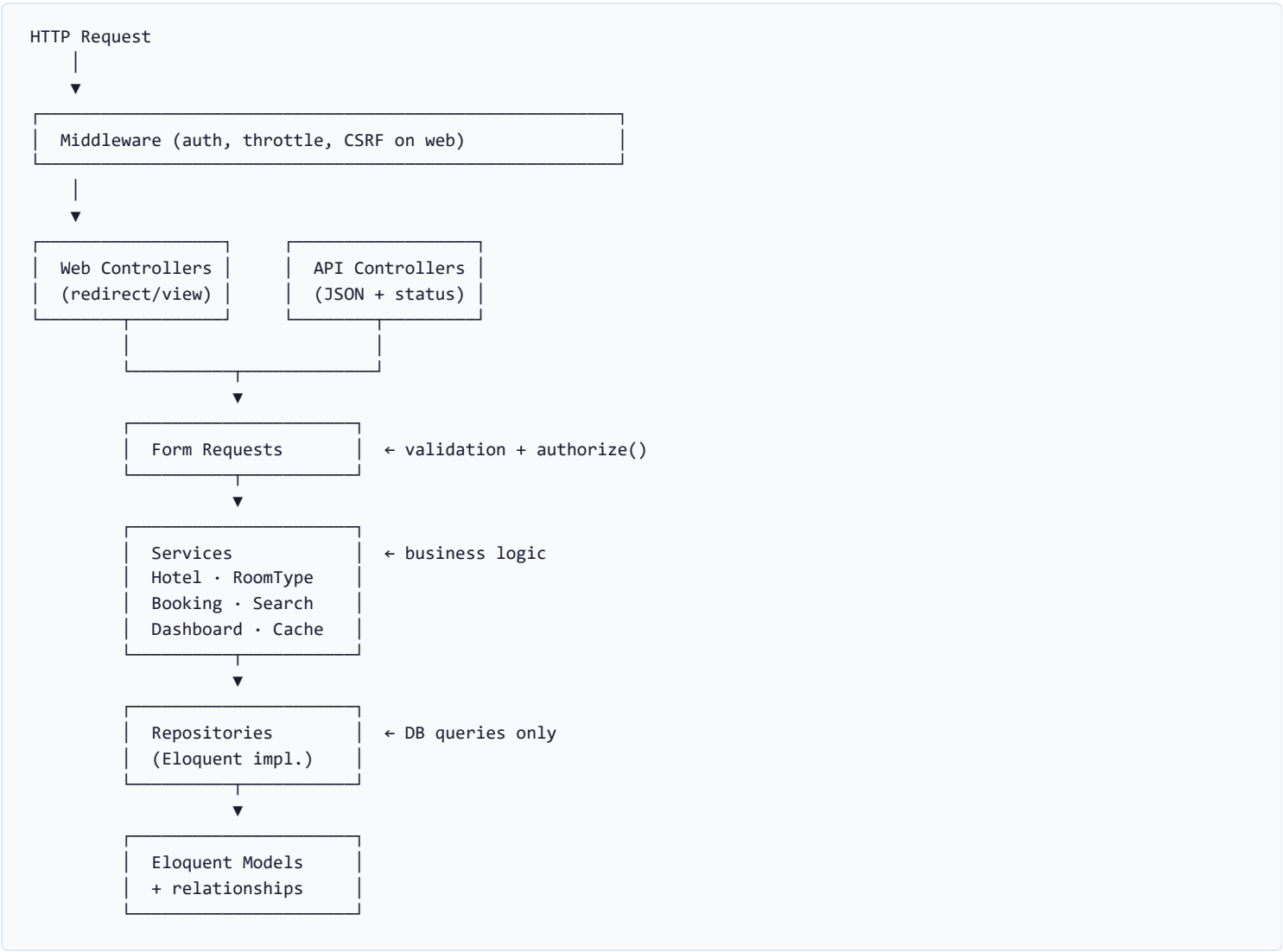
2. Brief Spec vs Implementation

Brief / spec idea	What we built	Notes
<code>rooms</code> table with <code>available_rooms</code> integer	<code>room_types</code> + <code>room_units</code> + <code>bookings</code>	Availability is computed per date range , not a static counter
Hotel CRUD	Web full CRUD + API list/create	API has no update/delete/show
Room CRUD	Web full CRUD + API create	Room = type + physical unit numbers
Search by city + dates	<code>SearchService</code> with overlap algorithm	Half-open interval <code>[checkin, checkout)</code>
Sanctum API auth	✓ Token create/revoke	Web uses session guard
Cache search results	✓ Redis, 60s TTL + version bump	Portable invalidation without cache tags
Docker	✓ Laravel Sail (PHP 8.3, MySQL 8, Redis)	<code>compose.yaml</code> + <code>sail.ps1</code> on Windows

Enhancements beyond the brief

Enhancement	Why it matters
UUID v4 primary keys	Safer public IDs; consistent across domain tables
Soft deletes	Hotels, room types, units, bookings recoverable
Audit columns	<code>created_by</code> , <code>updated_by</code> , <code>deleted_by</code> on all domain models
Spatie activity log	Full change history + <code>/activity</code> admin page
Repository pattern	Services depend on interfaces; Eloquent hidden behind contracts
Row-level locking	<code>lockForUpdate()</code> on booking create prevents double-booking
Countries / cities lookup	Seeded reference data for form dropdowns
Dashboard charts	Booking trends, occupancy %, top hotels/cities
Detail/show pages	Hotels, room types, bookings with audit panels
Rate limiting	Login, search, API group throttles
44 automated tests	Unit + feature coverage on SQLite in-memory

3. Architecture

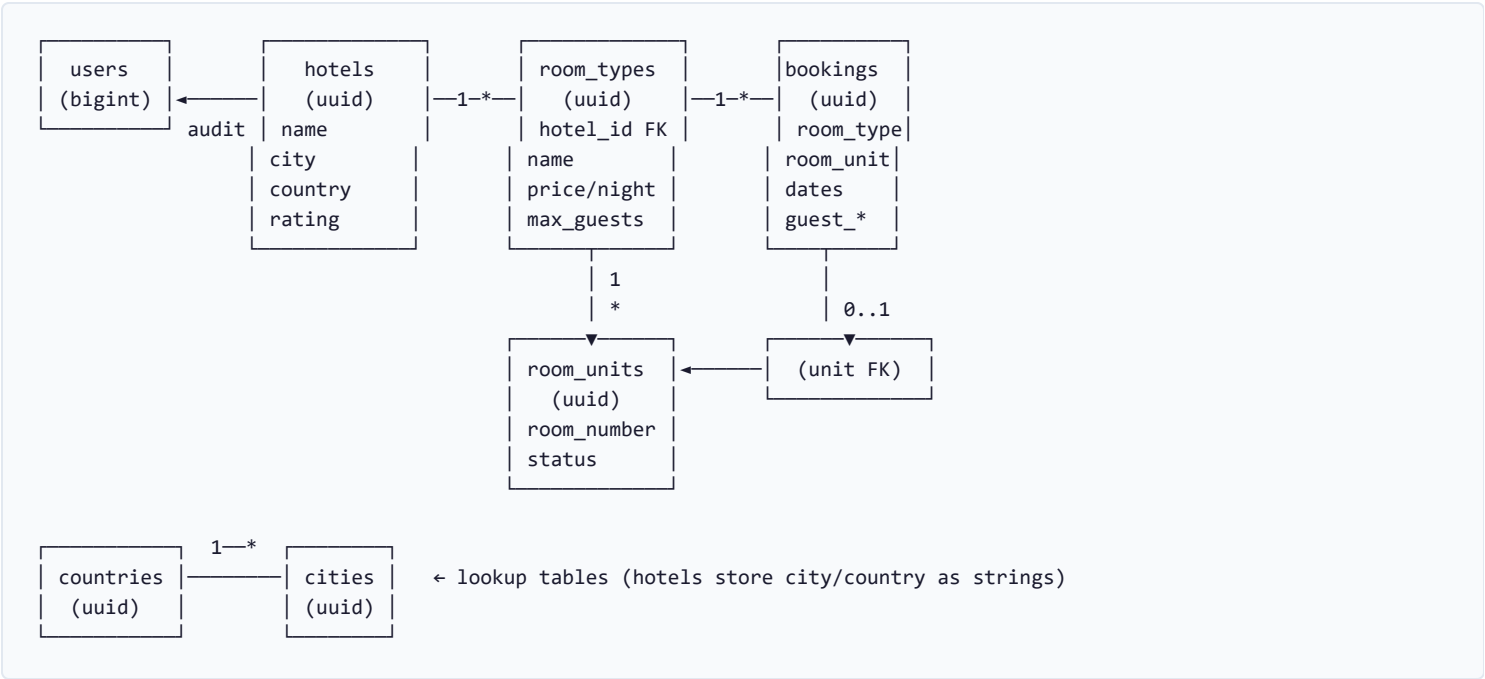


Key folders

Path	Purpose
app/Http/Controllers/Web/	Blade pages, redirects, flash messages
app/Http/Controllers/Api/	JSON responses, HTTP status codes
app/Http/Requests/	Validation rules
app/Http/Resources/	API response shape
app/Services/	Business logic
app/Repositories/	Data access
app/Models/AppModel.php	Base model: UUID, soft delete, audit, activity log

4. Database Design

4.1 Entity Relationship (core domain)



4.2 Table reference

hotels

Column	Type	Notes
id	UUID	PK
name	string	
city	string	indexed
country	string	
rating	tinyint 1-5	
created_by , updated_by , deleted_by	bigint nullable	→ users.id
deleted_at	timestamp	soft delete

Indexes: city , (city, rating)

room_types (migration: create_rooms_table)

Column	Type	Notes
id	UUID	PK
hotel_id	UUID	FK → hotels, cascade delete
name	string	e.g. "Deluxe King"
price_per_night	decimal(10,2)	
max_occupancy	smallint	
audit + soft delete		same pattern as hotels

room_units

Column	Type	Notes
id	UUID	PK
room_type_id	UUID	FK → room_types, cascade
room_number	string	e.g. "101", "102"
status	string	available · maintenance · blocked
audit + soft delete		

Unique: (room_type_id, room_number) · Index: (room_type_id, status)

bookings

Column	Type	Notes
id	UUID	PK
booking_reference	string	unique, e.g. BK-20260801-A3F2
room_type_id	UUID	FK → room_types
room_unit_id	UUID nullable	FK → room_units, null on delete
checkin_date / checkout_date	date	checkout day is free
guests	smallint	
guest_name , guest_email	string	
guest_phone	string nullable	
status	string	confirmed · cancelled
total_price	decimal(10,2)	snapshot at booking time
audit + soft delete		

Availability indexes:

- (room_type_id, status, checkin_date, checkout_date)
- (room_unit_id, status, checkin_date, checkout_date)

countries / cities

Reference data for dropdowns. Hotels store `city` and `country` as plain strings (denormalized for search simplicity).

cities	
country_id	FK → countries
name	unique per country

activity_log (Spatie)

Column	Purpose
subject_id/type	What changed (morph)
causer_id/type	Who changed it (morph → User)
event	created / updated / deleted
properties	JSON old/new values
log_name	grouping filter

users + personal_access_tokens

- `users` : standard Laravel auth (bigint PK)
- `personal_access_tokens` : Sanctum API tokens (morph to User)

4.3 Base model behaviour (AppModel)

All domain models extend `AppModel` :

1. **UUID v4** primary key (`HasVersion4Uuids`)
2. **Soft deletes** (`SoftDeletes`)
3. **Activity logging** — logs fillable attributes, dirty only
4. **Audit auto-stamp**: `created_by` / `updated_by` on save; `deleted_by` on soft delete

Relationships: `creator()` , `updater()` , `deleter()` → `User`

5. Application Flows

5.1 Web authentication (session)

```
Browser → GET /login → AuthController@showLogin → Blade form
Browser → POST /login → LoginRequest validates
    → Auth::attempt() → session regenerate
    → ActivityLogger "Logged in via web"
    → redirect /dashboard

POST /logout → ActivityLogger → Auth::logout()
    → session invalidate → redirect /login
```

- Middleware: `guest` on login, `auth` on all app pages
- CSRF protection on all POST/PUT/DELETE
- Rate limit: 5 attempts/min on login

5.2 API authentication (Sanctum)

```
POST /api/login → LoginRequest → Hash::check
    → 401 if fail
    → 200 { token, user } + activity log

Protected routes → Authorization: Bearer {token}
    → auth:sanctum middleware

POST /api/logout → revoke current token → 204
```

5.3 Hotel CRUD

Web: list (filters: country, city, rating) → create/edit modal → show detail page

API: GET /api/hotels (public, paginated) · POST /api/hotels (token, 201)

Delete guard: cannot delete hotel if it has room types (ResourceInUseException → 409)

```
HotelController → StoreHotelRequest → HotelService
    → HotelRepository → Hotel model
```

5.4 Room type CRUD

A "room" in the UI = **room type** (category) + **room units** (physical numbers).

```
StoreRoomTypeRequest → RoomTypeService::create()
    → insert room_types row
    → bulk insert room_units (parsed from "101, 102, 103")
    → SearchResultCache::bump()
```

Delete guard: cannot delete room type with existing bookings.

5.5 Search availability

```
SearchRequest (city, checkin, checkout, guests)
|
▼
SearchService::search()
|
├─ Build cache key: search:{version}:{md5(params)}
├─ Cache miss? Run query:
|   HotelRepository::availableInCity()
|   For each room type:
|       BookingService::availableUnitsForStay()
|       Filter by max_occupancy >= guests
|       Calculate nights × price_per_night
└─ Return hotels with available room types + unit numbers
```

Overlap rule (half-open [checkin, checkout]):

```
A unit is BLOCKED if a confirmed booking exists where:
existing.checkin_date < requested.checkout_date
AND existing.checkout_date > requested.checkin_date
```


- Cancelled bookings do **not** block availability
- Only units with `status = available` are searchable
- Checkout day is free for the next guest
- **Cache:** 60s TTL; inventory/booking changes bump version key

5.6 Booking create (overbooking-safe)

```
StoreBookingRequest → BookingService::create() [DB::transaction]
|
|→ RoomTypeRepository::findForUpdate()    ← row lock
|→ Validate guests <= max_occupancy
|→ resolveUnit():
|   If room_unit_id provided → lock that unit, verify free
|   Else → auto-pick first free unit for stay
|→ Generate reference: BK-{Ymd}-{RANDOM4}
|→ total_price = price_per_night × nights
|→ Insert booking (status: confirmed)
|→ SearchResultCache::bump()
```

Cancel: sets `status = cancelled` (does not hard-delete). Frees the unit for future searches.

5.7 Dashboard

```
DashboardController → aggregates:
- Hotel count, room type count, unit count
- Confirmed bookings, average rating
- Occupancy % (units with active confirmed stay today / total units)

DashboardService::charts() → JSON for Chart.js:
- 6-month booking + revenue trend
- Status doughnut (confirmed vs cancelled)
- Top 5 hotels by bookings
- Top 5 cities by hotel count
```

5.8 Activity log

- **Automatic:** Spatie `LogsActivity` on all `AppModel` children
 - **Manual:** `ActivityLogger` on web/API login and logout
 - **UI:** `GET /activity` — paginated log with filters
-

6. API Endpoints Summary

Method	Endpoint	Auth	Description
POST	/api/login	public	Returns { token, user }
POST	/api/logout	token	Revoke token (204)
GET	/api/hotels	public	Paginated list
POST	/api/hotels	token	Create hotel (201)
POST	/api/rooms	token	Create room type + units (201)
GET	/api/search	public	Availability search
GET	/api/room-types/{id}/available-units	public	Free units for dates
GET	/api/bookings	token	Paginated bookings
POST	/api/bookings	token	Create booking (201)
DELETE	/api/bookings/{id}	token	Cancel booking (204)

Rate limits: API group 60/min · login 10/min · search 30/min

7. Web Pages Summary

Route	Page
/login	Split-screen login with password toggle
/dashboard	Stats tiles + Chart.js charts
/hotels	List, filters, create/edit modal, show detail
/rooms	Room types list, create/edit, show detail
/bookings	Booking list, create form, show detail
/search	Date-aware availability search + results
/activity	Audit log viewer

8. Running the Project

Local (SQLite + Redis)

```
composer install && cp .env.example .env
php artisan key:generate && php artisan migrate:fresh --seed
php artisan serve    # http://127.0.0.1:8000
```

Windows: `.\run-local.ps1 up`

Docker (Laravel Sail)

```
# merge .env.docker.example into .env
.\sail.ps1 build && .\sail.ps1 setup    # Windows
# http://localhost:8080/login
```

Tests

```
php artisan test          # 44 tests, SQLite in-memory
.\sail.ps1 test           # same suite inside container
```

9. Test Coverage Map

Test file	What it verifies
AvailabilityTest (unit)	Overlap logic, checkout-day-free rule
SearchAvailabilityTest	Search results, pricing, guest filter
BookingTest	Create, 422 when full, cache invalidation, cancel
AuthTest	Web session + API token flows
InventoryTest	Delete guards, API city filter
AuditAndSoftDeleteTest	UUID, audit columns, soft delete
ActivityLogTest	Spatie logging on CRUD + auth
ViewPagesTest	Show pages, dashboard charts, auth gate

10. File Quick Reference

Concern	Primary files
Search + cache	<code>app/Services/SearchService.php</code> , <code>app/Services/Search/SearchResultCache.php</code>
Booking logic	<code>app/Services/BookingService.php</code>
Availability query	<code>app/Repositories/Eloquent/EloquentRoomUnitRepository.php</code>
API errors	<code>app/Exceptions/ApiExceptionRenderer.php</code>
Repo bindings	<code>app/Providers/RepositoryServiceProvider.php</code>
Sail Docker	<code>compose.yaml</code> , <code>sail.ps1</code>
Seed data	<code>database/seeders/LocationSeeder.php</code> , <code>DatabaseSeeder.php</code>

Generated for TerraStay / hotel-booking-temp — Laravel 13 interview submission.